11-1b Cooperation and Collusion

In *The Wealth of Nations* (1776), Adam Smith wrote: "People of the same trade seldom meet together, even for merriment and diversion, but their conversation often ends in a conspiracy against the public." In modern jargon, this means that competing firms in an industry may have an incentive to engage in **collusion** (Collective attempts between competing firms to reduce competition.) , defined as collective attempts to reduce competition.

Because managers (and students) generally do not like to discuss "collusion," another "C" word, coordination, is now frequently used in preference to collusion. However, given the legal battles centered on collusion, managers (and students) cannot shy away from it. Instead, they need to confront the legal definitions and debates about collusion. Collusion can be tacit or explicit. Firms engage in **tacit colluion** (Firms indirectly coordinate actions by signaling their intention to reduce output and maintain pricing above competitive levels.) when they *indirectly* coordinate actions by signaling their intention to reduce output and maintain pricing above competitive levels. **Explicit collusion** (Firms directly negotiate output and pricing and divide markets.) exists when firms *directly* negotiate output and pricing and divide markets. Explicit collusion leads to a **cartel** (An output-fixing and price-fixing entity involving multiple competitors.) —an output-fixing and price-fixing entity involving multiple competitors. A cartel is also known as a trust, whose members have to trust each other in honoring agreements. Since the Sherman Act of 1890, cartels have often been labeled "anticompetitive" and outlawed by **antitrust laws** (Law that makes cartels (trusts) illegal.) in many countries.

In addition to antitrust laws, collusion often suffers from a **prisoners' dilemma** (In game theory, a type of game in which the outcome depends on two parties deciding whether to cooperate or to defect.) , which underpins **game theory** (A theory that studies the interactions between two parties that compete and/or cooperate with each other.) . The term "prisoners' dilemma" derives from a simple game in which two prisoners suspected of a major joint crime (such as burglary) are separately interrogated and told that if either one confesses, the confessor will get a one-year sentence, while the other will go to jail for ten years. Since the police do not have strong incriminating evidence for the more serious burglary charges, if neither confesses, both will be convicted of a lesser charge (such as trespassing), each for two years. If both confess, both will go to jail for ten years. At a first glance, the solution to this problem seems clear enough. The maximum *joint* payoff would be for neither of them to confess. However, even if both parties agree not to confess before they are arrested, there are still tremendous incentives to confess.

Translated to an airline setting, Figure 11.1 illustrates the payoff structure for both airlines A and B in a given market—let's say, between Dubai and Cairo. Assuming a total of 200 passengers, Cell 1 represents the most ideal outcome for both airlines to maintain the price at US$500, and each gets 100 passengers and makes US$50,000—the "industry" revenue reaches US$100,000. In Cell 2, if B maintains its price at US$500 while A drops it to US$300, B is likely to lose all customers. Assuming perfectly transparent pricing information on the Internet, who would want to pay US$500 when you can get a ticket for US$300? Thus, A may make US$60,000 on 200 passengers while B gets nobody. In Cell 3, the situation is reversed. In both Cells 2 and 3, although the industry *decreases* revenue by 40%, the price dropper *increases* its own revenue by 20%. Thus, both A and B have strong incentives to reduce price and hope the other side to become a "sucker." However, neither likes to be a "sucker." Thus, both A and B may want to chop prices, as in Cell 4, whereby each still gets 100 passengers. But both firms, as well as the industry, end up with a 40% reduction of revenue. A key insight of game theory is that even if A and B have a prior agreement to fix the price at US$500, both still have strong incentives to cheat, thus pulling the industry to Cell 4 whereby both are clearly worse off.

Figure 11.1

A "Prisoners' Dilemma" for Airlines and Payoff Structure (assuming a total of 200 passengers in one market consisting of two cities)

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